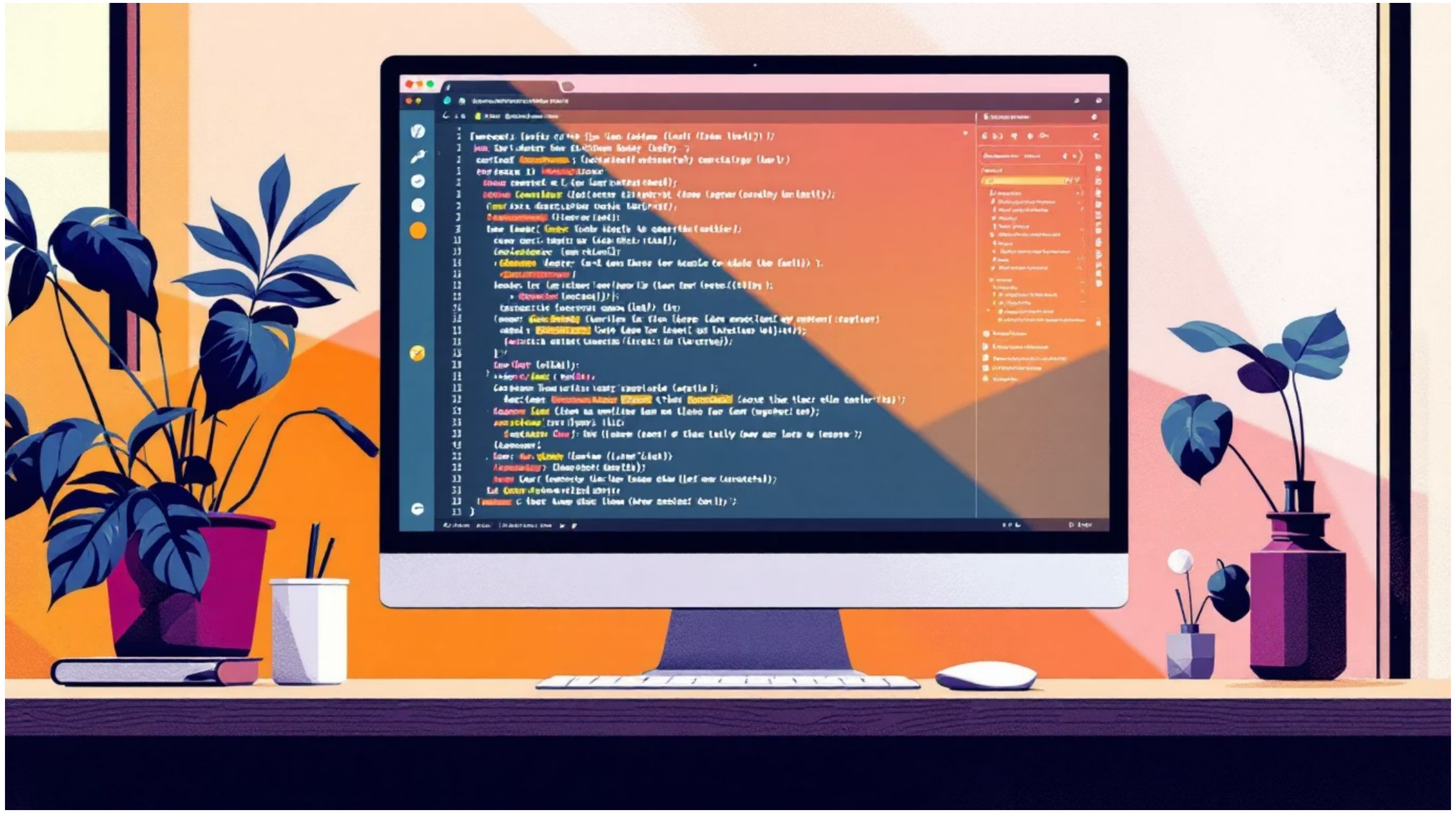
Mastering Text-to-SQL with Defog SQL Coder

Overcoming inaccurate queries through comprehensive schema context and advanced prompt engineering techniques.

Defog SQL Coder: The Specialized Solution

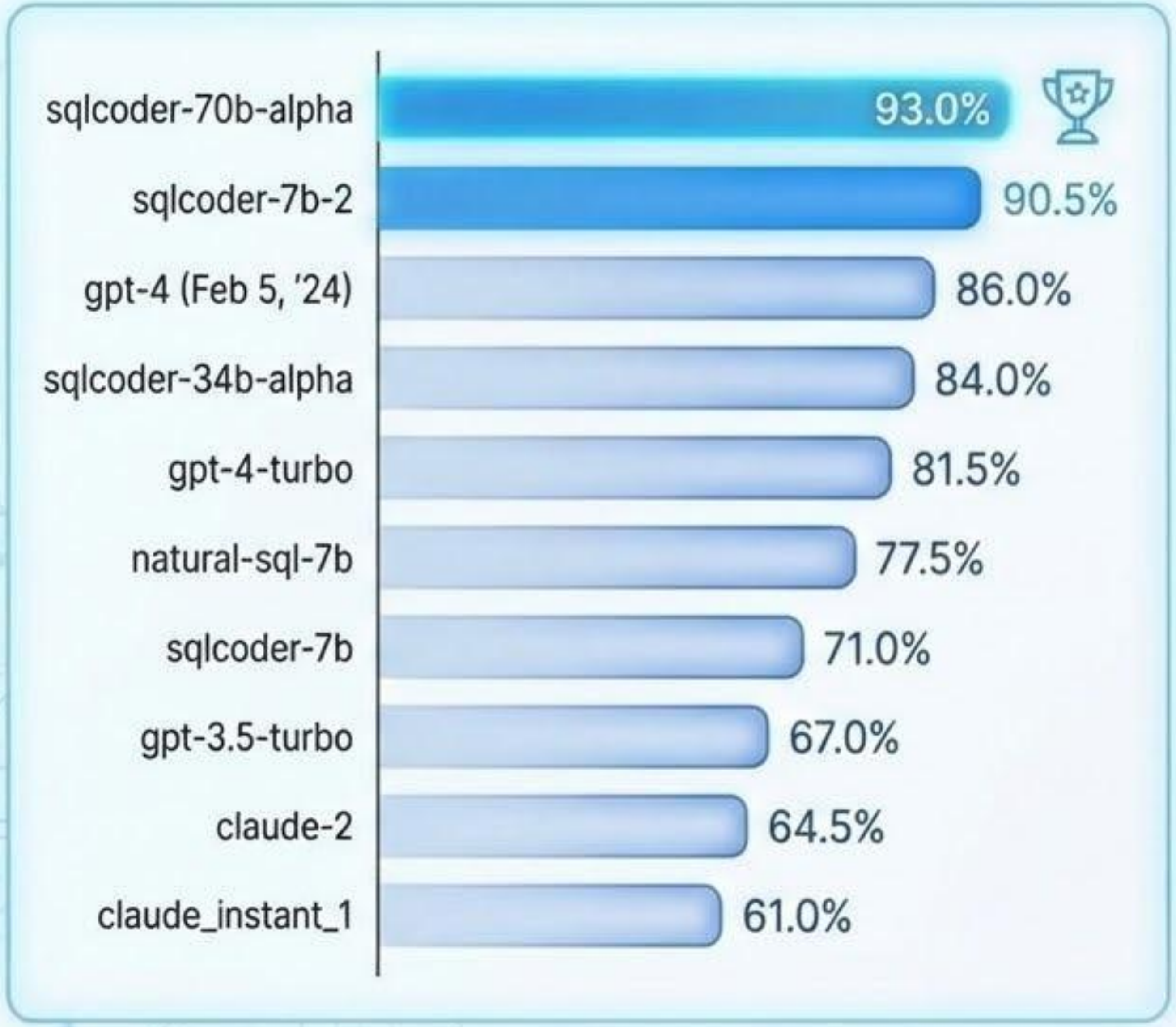


An open-source, state-of-the-art LLM fine-tuned specifically for converting natural language to SQL queries.

Hardware Requirements

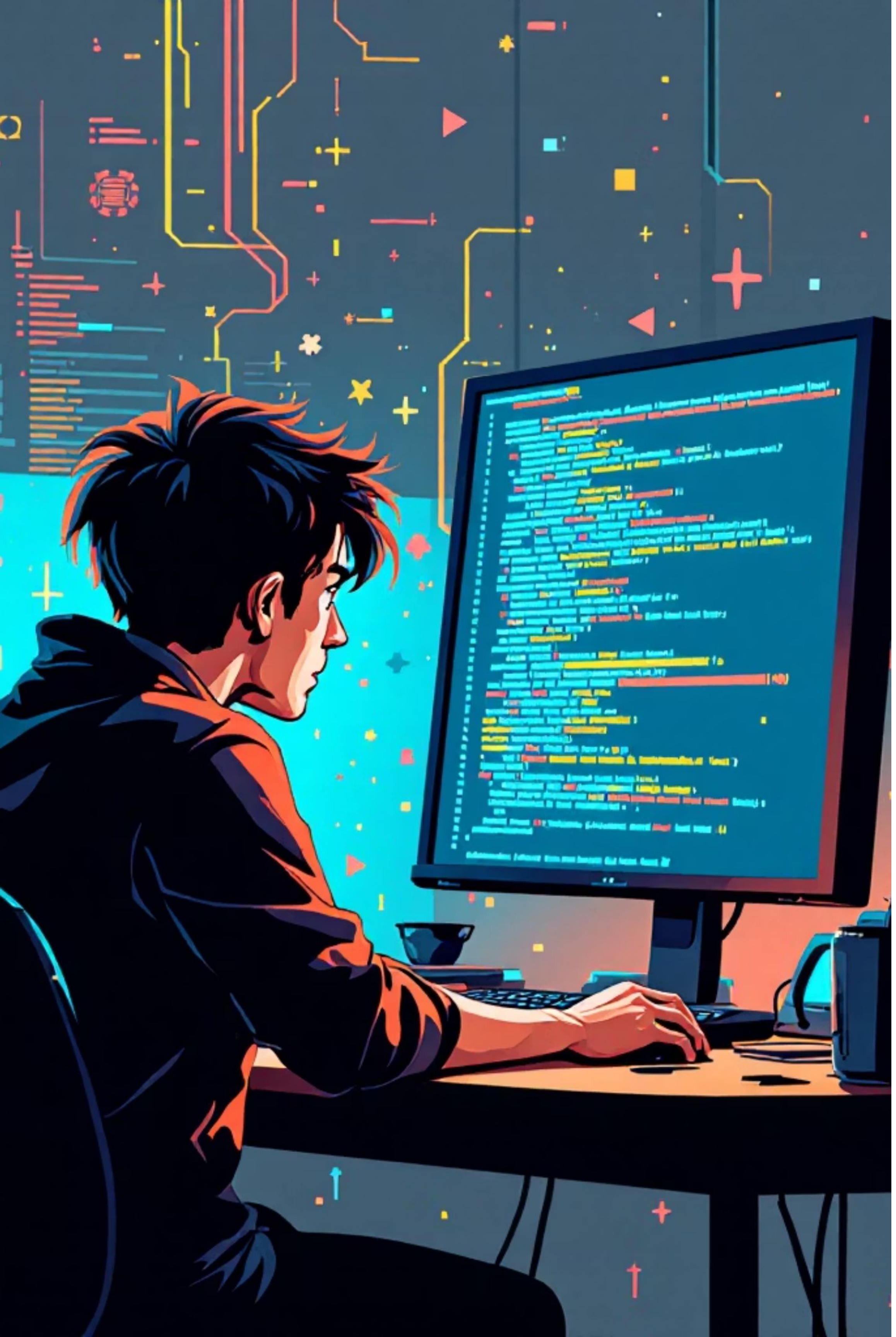
- GPU VRAM: 16GB+ for 15B model
- Quantization: 4-bit/8-bit for smaller GPUs
- Software stack: Hugging Face Transformers, PyTorch

Why Defog Outperforms General Models



Percentage of correctly generated SQL queries on novel schemas (n=200)

- sqlcoder-70b-alpha leads with 93.0% accuracy on novel schemas
- Defog models (70b-alpha, 70b-2, 34b-alpha) consistently outperform general-purpose models
- GPT-4 achieves 86.0% accuracy, demonstrating the gap between specialized and general models
- Defog's domain-specific training translates to superior real-world performance
- Specialized fine-tuning beats larger general models when accuracy matters most



The Initial Roadblock

The Problem

Model generated highly inaccurate and illogical SQL queries despite powerful architecture

Root Cause

Basic prompt lacked proper database schema specification and strict guiding rules

Diagnosing the Failure

The Investigation

Results weren't just slightly off—they were completely unusable. The model hallucinated relationships and guessed column names.

> "The model was flying blind. It knew SQL syntax perfectly, but had zero context about our database's specific shape and contents."

COLUMN NAME	DATA TYPE	SAMPLE DATA / UNIQUE VALUES
element	text	node, relation, way
id	bigint	3318424452, 5372466, 368848142, 368851385, 4511596613
geometry	USER-DEFINED	0101000020E6100000CCCF004ĐD9E3, 0106000020E0100000019
air_conditioning	text	yes, no
amenity	text	restaurant, blood_bank, cafe, club, clinic, research_
created_by	text	Potlatch 0,100
name	text	VGS mess, Malviya Badminton Court, C1fs cofonference_
opening_hours	text	Potlatch 01
check_date	text	2025-03-29, 2025:54-11, 2025-05-02, 2025-05-
tourism	text	2025-06-13, hin 2
brand	text	Indian Oil, Looters, Bharat Petroleum, Da
brand:wikidata	text	Q854626, Q1298348
shop	text	hairdresser, supernarket, department_store, convenien
healthcare	text	butlery, pharmasy
building	text	university=department, dormitory, apartments, residen
bulding:levels	text	1, 2, 4, 3
religion	text	hindu
sport	text	nature_reserve, pitch, park, garden, sports_centre, s
sport	text	smimming_pool, playground, hateaway
man_made	text	sllo
addr:district	text	Pilani, Jhunjhumu
addr:street	text	333031, 233031, 306401
addr:Ststeet	text	Krishna Marg, Ganesh Colony, Vidya Vihar, Gandhi Marg
addr:city	text	Pilani
addr:full	text	Pilani, Jhunjhunu, Jhunjhumu, Bharat Colony, Vètia:Pial
apcr:city	text	10005285633
plani	text	233031, 213360401
addr:postcode	text	Krishna Marg, Pilani1, 01505505508, +91982658499,
addr:street	text	siumpesal.biralton.Cnphotnailipal.com, गिलांस-नीवर्षी वास्पा वेवीनिवारे के ला पा
email	text	opnmonaaem.singh@openodata.com
email	text	10002585533
air_continue	text	10005285333, 01569742288
operator	text	5295005840, +30 e1, 8910526588.1Q9,
room	text	https://hww.bitw-pilni.as.in/
email	text	18003285323
contater:phone	text	180055853333,a, Jhunjhunu, +995805056, 819Γl0265nS0G
contact:mobile	text	1590242960, 910881
room:en	text	cptemccedge singh3
opcat0rtype	text	18005285333, 91569242288,
phone	text	18005283335
loiqure	text	yes, no
webs	text	howelnowungo, singh, ernpis, 939vq31dl3ad
wikidsata	text	texe
histori	text	Pilani Jhunjhunu, Jhunjhumu, Bharat Colony,
operator:type	text	vedhwal, gine:
contact:phone	text	Krishna Marg, Ganesh Colony, Vidya Vihar, Gandhi Marg
website	text	Pilani, Jhunjhumu
website:mobsite	text	badminton, cricket, table_tenni, multi, volleyball,
email	text	skateboard
man:made	text	statue
hemal	text	nP+
brand:wikidata	text	Pilai, Jhunjhunu H
	text	sllo
room	text	16005285333, 10180?263 09590595880,
air_contimue	text	peine, ngo, university
email	text	yes no
operator	text	OpenGnowungo, university
office:-type	text	private, ngo, university, h040Fsनोर्ड: आमवारफ्स वीएनी ओरे पृ
email	text	jes no
contact :feahe	text	Pilani, JHhunJMumu, Ricco Industrial Area, Poonia 3,
wikidata	text	Rojasthan, Biasaita

Solution: Comprehensive Schema Injection

01

Extract All Columns

Complete database structure mapped

02

Capture Value Context

Distinct values for each column

03

Build Mega-Prompt

Full schema as base context

This overhaul gave the model full visibility into our database structure, eliminating guesswork.

Solution: Custom Logic Rules

Enhanced System Rules

Strict boundaries on query writing and JOIN operations injected into prompts.

Data Mapping Reference

Bridged gap between human language and database structure with specific mappings.

> **Key Insight:** Even fine-tuned models need explicit business logic to handle real-world data complexity.

Fixing the "AND/OR" Logic Failure

1

The Problem

Query used AND for mutually exclusive conditions, creating impossible logic

2

The Fix

Engineered strict tokenization rules into prompt

3

The Result

Valid queries using OR combinations

Tokenization Rules Implemented

1. Split user phrase into meaningful tokens
2. Ignore filler words: on campus, in Pilani, at BITS, near, inside
3. For multiple tokens, combine using OR
4. Never use AND between name ILIKE conditions
5. Never require full phrase match unless explicitly stated

Real-World Example: Meera Bhawan Query

Human Query : Find 'Meera Bhawan' (Girls Hostel).

The Problem Query

```
SELECT p.name FROM pilani_places p WHERE p.name ILIKE '%Meera Bhawan%' AND p.name ILIKE '%Girls
Hostel%';
```

This query is impossible—no single name can match both conditions simultaneously.

The Corrected Query

```
SELECT p.name FROM pilani_places p WHERE (p.name ILIKE '%Meera Bhawan%' OR p.name ILIKE '%Girls
Hostel%');
```

By tokenizing the input and combining with OR instead of AND, the model now returns valid results.

Final Results & Conclusion

High

Accuracy

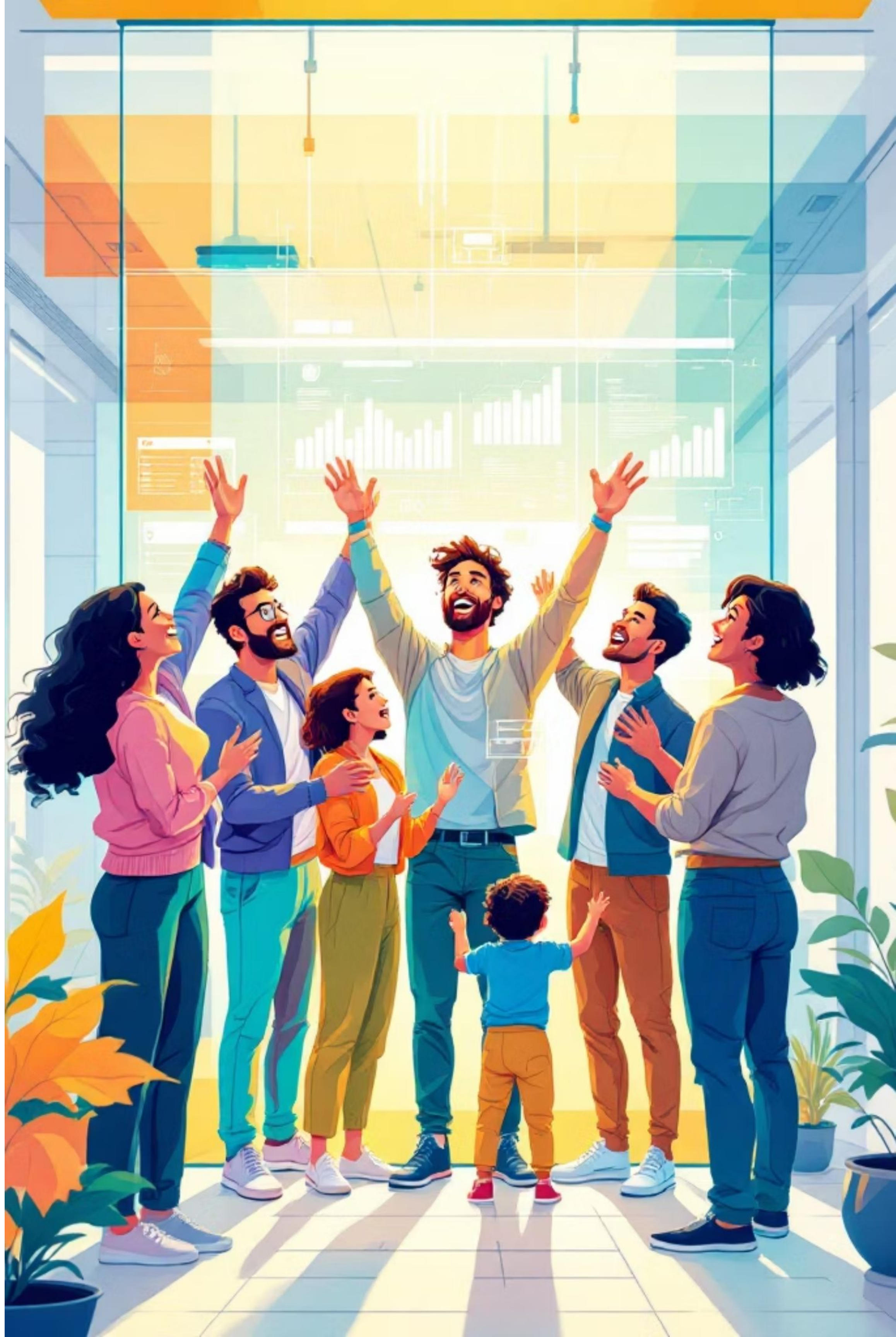
Queries now reliably executable

0

Hallucinations

No more guessed relationships

> Even specialized, fine-tuned models require rigorous prompt engineering, full schema awareness, and specific business rules to handle real-world data effectively.



Challenges & Path Forward

1

2

Environment & Hardware

Dependency conflicts and complex library requirements on IPU GPU cluster. Still working through configuration issues.

Workflow Bottlenecks

Spent significant time debugging in isolation. Need guidance on some problems.

3

Mentorship Request

Requesting Research Scholar as point of contact to navigate complex technical blockers more efficiently.